



Due-diligence and research-security tradecraft, built against real open data.

— A SELF-DIRECTED, AI-ASSISTED
SOFTWARE ENGINEERING PROJECT —

Overview

A public, non-classified portfolio project that reproduces — at hobby scale, against real open data — the core capability behind enterprise due-diligence platforms, threat and risk screening, and bibliometric affiliation matching.

01 Entity resolution across multiple heterogeneous real datasets — not one clean list.

02 Confidence-scored, evidence-traceable output an auditor can follow back to source.

03 Foreign-ownership and affiliation screening, including a bibliometric co-authorship layer.

04 “Candidate match, not confirmed finding” built into the data model itself — not just stated in a README.

Integrated Data Resources

NSF Award Search

Public federal award records — the entity and subject source for the batch-screening path.

DoD Section 1260H List

Chinese military-linked companies, published per federal statute.

Dept. of Education §117 Disclosures

Foreign gift and contract records, cross-checked against declared affiliations.

Federal critical-technology lists

Secondary corpus for semantic topic-matching against flagged technology domains.

OpenSanctions (consolidated)

Restricted-party and entity-of-concern screening across OFAC, BIS, and State Department lists.

SDN · Denied Persons · Entity · Unverified · Military
End-User · AECA Debarred · Nonproliferation · FTO

GLEIF Golden Copy

Global corporate ownership data — traces parent/subsidiary chains and flags foreign-controlled entities.

OpenAlex

Global bibliometric graph of publications, authorship, and institutional affiliation — surfaces co-authorship ties an institution's own records may miss.

Engineering & Architecture

SERVICE LAYER

A REST API (FastAPI) separating the core engine from the UI.

DEPLOYMENT

Containerized (Docker), served in production behind a reverse proxy.

ENTITY RESOLUTION

Fuzzy and confidence-scored — no component outputs a bare true/false match.

INFRASTRUCTURE

Infrastructure-as-code (Terraform) for reproducible cloud provisioning.

ANALYTICAL STORE

An embedded columnar database for large, join-heavy workloads — no separate DB server.

SECURITY

Defense in depth: server-side path allowlisting, gated mutating actions, rate limiting.

CONTINUOUS INTEGRATION

An automated test suite run on every change.

SEMANTIC MATCHING

Vector-similarity search over technical topics, with a version-pinned model for reproducibility.

REPRODUCIBILITY

Every run records exactly which data snapshot and code version produced it.

Use Case & Workflow

Subject in, documented decision out — never a black-box risk score.

Case-based compliance review

A triggering event — a new hire, a visiting researcher — opens a case. The system reconciles what the person declared against what public records show.

Discrepancies surface as observable facts — no risk score, no severity label.

An analyst reviews and records a decision with a documented reason.

The case file accumulates into a defensible, auditable record.

Restricted-party screening

A hire, a visiting-scholar invitation, or a purchasing or financial transaction triggers a name screen against government restricted-party lists.

Every candidate match is confidence-scored and linked to its evidence.

An analyst disposes each one — dismiss or escalate.

Each disposition carries a documented reason code.

A hands-on demonstration of applied engineering and compliance-domain tradecraft.

NOT a production tool, and not legal or compliance advice.